

DYNA METFORMIN TABLET 500MG

MAL19961876AZ

DESCRIPTION:

TABLET

Shape : Round/Flat/Scored

Colour : White

Marking : "M"

Oral Hypoglycaemic

EACH TABLET CONTAINS:

Metformin HCl 500 mg

PHARMACODYNAMICS:

Metformin's mode of action is not clear. It does not stimulate Insulin releases but does require some Insulin be present for it to exert a hypoglycaemic effect. Possible mechanisms of action include delay in the absorption of Glucose from the gastrointestinal tract, an increase in Insulin sensitivity, and inhibition of hepatic gluconeogenesis.

PHARMACOKINETICS:

Metformin Hydrochloride is absorbed from the gastrointestinal tract and is excreted, unchanged, in the urine. It has a plasma half-life of about 3 hours and it is not bound to plasma proteins.

INDICATION:

Control of hyperglycaemic of maturity onset diabetes mellitus. It is used in non Insulin-dependant diabetes mellitus when diet control fails. It is sometimes given to patients no longer responding to the sulphonylureas.

RECOMMENDED DOSE:

Initial dose of 500 mg three times daily or 850 mg twice daily with meals, gradually increased if necessary to a maximum of 3 g daily.

Renal impairment:

A glomerular filtration rate (GFR) should be assessed before initiation of treatment with Metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

GFR mL/min	Total maximum daily dose (to be divided into 2-3 daily doses)	Additional considerations
60-89	3000 mg	Dose reduction may be considered in relation to declining renal function.
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of Metformin. The starting dose is at most half of the maximum dose.
30-44	1000 mg	
<30	-	Metformin is contraindicated

ROUTE OF ADMINISTRATION:

FOR ORAL USE ONLY

CONTRAINDICATIONS:

Metformin should not be used in Insulin-dependant diabetes mellitus.

Use in non Insulin-dependant diabetes mellitus is contraindicated in patients with ketoacidosis and in those with severe infection, stress, trauma, or other severe conditions where the sulphonylurea is unlikely to control the hyperglycaemia; Insulin should be administered in such situations.

Metformin should not be given in severe impairment of renal or hepatic function because of an increased risk of hypoglycaemia. It is recommended that Metformin not be used in patients with severe impairment of thyroid function. Debilitated; malnourished; and elderly patients are particularly susceptible to the hypoglycaemic effects of sulphonylureas and some authorities recommend that Metformin with its long half-life be avoided in such patients.

Care is necessary during excessive exercise as hypoglycaemia may be provoked.

Metformin antidiuretic effect may cause problems in patients with conditions associated with fluid retention.

Metformin is excreted in breast-milk and patients taking Metformin should not breast feed.

Metformin should not be used in patients with heart failure, dehydration, acute or chronic alcoholism, or any other conditions likely to predispose to lactic acidosis.

- Severely reduced kidney function (GFR <30 mL/min)
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)

WARNINGS AND PRECAUTIONS:

It is excreted by the kidney and regular monitoring of renal function is advised in all diabetics. The use of Metformin is not advised in conditions which may cause dehydration or in patients suffering from serious infections or trauma. Patients receiving continuous Metformin therapy should have an annual estimation of Vitamin B₁₂ levels because of reports of decreased Vitamin B₁₂ absorption. Blood glucose should be monitored because combined therapy with a sulphonylurea may cause hypoglycaemia.

Use in children: Not recommended.

Lactic acidosis

Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (< 7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal function

GFR should be assessed before treatment initiation and regularly thereafter [See Section **Recommended Dosage**]. Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function [See Section **Contraindications**].

Metformin may reduce vitamin B₁₂ serum levels. The risk of low vitamin B₁₂ levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B₁₂ deficiency. In case of suspicion of vitamin B₁₂ deficiency (such as anemia or neuropathy), vitamin B₁₂ serum levels should be monitored. Periodic vitamin B₁₂ monitoring could be necessary in patients with risk factors for vitamin B₁₂ deficiency. Metformin therapy should be continued for as long as it is tolerated and not contraindicated and appropriate corrective treatment for vitamin B₁₂ deficiency provided in line with current clinical guidelines.

DRUG INTERACTIONS:

An interaction between Metformin and anticoagulants is a possibility and dosage of the latter may need adjustment.

PREGNANCY AND LACTATION:

Use in pregnancy: The use of Metformin is not advisable.

Use in lactation: No information is available.

SIDE EFFECTS:

Metformin causes gastrointestinal adverse effects with anorexia, nausea, and vomiting, absorption of various substances including Vitamin B₁₂ may be impaired. Patients may experience a metallic taste and there may be weight loss. Metformin may also cause lactic acidosis.

Metabolism and nutrition disorders

Common: Vitamin B₁₂ decrease/deficiency

SYMPTOM AND TREATMENT OF OVERDOSE:

Overdose may cause lactic acidosis, hypoglycaemia or gastrointestinal discomfort with anorexia, nausea, and vomiting. Acute poisoning with Metformin calls for intensive supportive therapy. Lactic acidosis may require treatment with Sodium Bicarbonate as well as other measures. Glucose and glucagon may be required for hypoglycaemia.

PACKING/PACK SIZE(S):

Blister pack of 10x10's and 100x10's.

**JAUHI UBAT DARIPADA KANAK-KANAK
KEEP OUT OF REACH OF CHILDREN**

MANUFACTURER/PRODUCT REGISTRATION HOLDER:

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